Week 4 Day 2 – Core Supervised Learning

Preprocessing Choices

The Adult Income dataset contains both numeric and categorical features. I used a **ColumnTransformer** to preprocess both types of data separately.

For numeric features, I used **SimpleImputer (median)** to fill missing values because it is less affected by outliers. Then I applied **StandardScaler** to scale the numeric data.

For categorical features, I used **SimpleImputer (most_frequent)** to fill missing values and **OneHotEncoder** to convert text data into numbers. Using a pipeline also prevents **data leakage** because preprocessing is done only on the training data.

Model Comparison

Three models were compared: Rule-Based Baseline, Logistic Regression, and Decision Tree.

Metric	Rule Baseline	Logistic Regression	Decision Tree
Accuracy	0.752994	**0.852083**	0.820964
Precision	0.484368	**0.739667**	0.625909
Recall	0.497006	0.589393	**0.626176**
F1 Score	0.490606	**0.656034**	0.626042
ROC AUC	0.665271	**0.903183**	0.754402
PR AUC	0.361115	**0.761198**	0.481561

Logistic Regression gave the best overall performance. It achieved the highest Accuracy, Precision, F1 Score, ROC AUC, and PR AUC. The Decision Tree had slightly higher Recall but showed overfitting because its training accuracy was much higher than its testing accuracy.

Model Selection

I will continue with **Logistic Regression** because it gives the highest Precision, which is the main goal of this project. It also performs better than the Decision Tree on most evaluation metrics and generalizes better on unseen data.

Future Work

The preprocessing pipeline will be reused in future experiments. Next, I will tune the model parameters, try more feature engineering, and improve the model while keeping **Precision** as the main evaluation metric.